

Test Plan

Team 13

MEDBOOK

TEST PLAN

Date: 07/12/2026

Introduction

The Test Plan has been created to communicate the test approach to team members. It includes the objectives, scope, schedule, risks, and approach. This document will clearly identify what the test deliverables will be and what is deemed in and out of scope.

1.1 Objectives

The Med Book Appointment System was developed to aid physicians and their patients in booking appointments electronically. Using our web and mobile app, patients can easily sign up, look up providers, book appointments, cancel appointments, reschedule appointments, and set reminders for appointments. Physicians are also able to manage their calendars by looking at patient appointments and easily adding or blocking their availability. This system purposes to improve communication between patients/providers and allow for better scheduling of medical appointments. Benefits of using the Med Book Appointment System include time savings, decreasing the number of missed appointments, and an enhanced patient experience.

Sprint 1 will consist of developing the minimum features needed to start up the Med Book Appointment System. Those features include:

- Register/Create user account
- Login/authentication
- Create/edit Patient Info
- Doctor search/Provider Directory
- Making an Appointment
- Viewing Appointments
- Canceling/Appointment changing

These features were chosen because they are considered the absolute minimum the system needs to function. There needs to be a capability to add advanced features, such as appointment reminders, paying online, messaging, and adding medical records. However, before a patient can use those features, they need to be able to create an account, find a provider, and book an appointment. Completing these features in Sprint 1 lets the team get a large portion of the program done and test it sooner, so end users can give feedback on what they want changed or added in future sprints.

1.2 Team Members

Resource Name	Role
Hamidat Oloko	Team lead and Support
Nesna Prasai	Documentation
Asif Khan	Backend
Genesis Enemuoh	Front end

2 Scope

The following sections indicate what is tested during each sprint. The scope of testing is determined at the beginning of the current sprint.

At the end of Sprint 1 a user must be able to:

- Create a patient account.
- Log in with valid credentials.
- Create or edit patient information.
- Search for healthcare providers.
- Schedule an appointment.
- View scheduled appointments.
- Cancel or reschedule appointments.
- Navigate between the main application screens.

Other testing performed during Sprint 1:

- Functional Testing
- User Interface Testing
- Input Validation Testing
- Navigation Testing
- Usability Testing

At the end of Sprint 2, a user must be able to:

- View provider information.
- View appointment details.
- Edit appointments.
- Review appointment history.
- Verify navigation between all prototype pages.

Other testing performed during Sprint 2:

- Functional Testing
- Integration Testing between prototype screens
- Regression Testing
- Responsive Layout Testing

At the end of Sprint 3, a user must be able to:

- Access patient screens.
- Access administrator screens.
- View role-specific features.
- Complete the appointment workflow without navigation errors.
- Use the prototype on desktop and mobile layouts.

Other testing performed during Sprint 3:

- User Acceptance Testing
- Role-Based Access Testing
- Interface Consistency Testing
- Security Review

- Final Prototype Validation

3 Assumptions / Risks

3.1 Assumptions

The following assumptions apply to the MEDBOOK Appointment System:

- Users have access to a stable internet connection.
- Users access the system using a supported web browser or mobile device.
- The MEDBOOK prototype accurately represents the planned functionality of the final application.
- Sample patient and appointment information will be used during testing.
- Separate patient and administrator accounts are available for testing different user roles.
- Team members complete their assigned prototype screens before testing begins.
- All navigation links between prototype screens are functional before testing starts.

3.2 Risks

The following risks have been identified and the appropriate action identified to mitigate their impact on the project. The impact (or severity) of the risk is based on how the project would be affected if the risk was triggered. The trigger is what milestone or event would cause the risk to become an issue to be dealt with.

#	Risk	Impact	Trigger	Mitigation Plan
1	Scope creep - as testers become more familiar with the tool, they will want more functionality	High	Delays in implementation date	Each iteration, functionality will be closely monitored. Priorities will be set and discussed by stakeholders. Since the driver is functionality and not time, it may be necessary to push the date out.
2	Weekly delivery is not possible because the developer works off site	Medium	Product did not get delivered on schedule	Members will communicate regularly, monitor progress during each sprint, and allocate work when necessary to keep development on schedule.
3	Users become confused due to inconsistent screen layout	Medium	Different pages use different layout	Maintain a consistent interface design and review all screens before testing.
4	Navigation links between prototype screens do not work correctly	Medium	Users cannot continue through the appointment process	Verify every navigation link during testing and correct the broken connection before reviewing.
5	Appointment workflow is incomplete	High	Missing screens / navigation	Review the complete workflow as a team and validate it before final submission.

#	Risk	Impact	Trigger	Mitigation Plan
6	Team members are unfamiliar with manual test-case documentation for a Figma prototype	Low	Test cases are inconsistent or missing steps	Use a shared test-case template and have the team lead review each set of test cases before execution.

4 Test Approach

The project is using an agile approach, with 2-week sprints.

Sprint 3

During Sprint 3, the team will perform functional testing to verify that the prototype meets the project requirements. Testing will focus on user registration, login, provider search, appointment scheduling, appointment management, and navigation between screens. Manual testing will be used to confirm that each feature behaves as expected and that all user interface elements function correctly.

Sprint 4

During Sprint 4, integration testing and regression testing will be performed. The team will verify that all prototype screens connect correctly, and that updates made during development do not affect previously completed features. Testing will also include usability checks to ensure the application is easy to navigate.

Sprint 5

Sprint 5 will focus on final system validation. User acceptance testing (UAT), role-based access testing, interface consistency testing, and final usability testing will be completed. Team members will review the entire prototype to ensure it satisfies the project requirements before submission.

4.1 Test Automation

Since MEDBOOK is currently a Figma prototype, testing will primarily be performed manually. Team members will test navigation, user interface consistency, usability, and the appointment scheduling workflow. Manual testing allows the team to verify that users can complete common tasks successfully and that all prototype screens are connected correctly.

Automated testing tools may be introduced in future development once the system is implemented as a working application. At the current stage, manual testing is the most appropriate approach because there is no executable code to automate.

Test Cases (Black Box)

User Registration and Login

Test Case ID	Description	Requirements Trace	Directions	Expected Output	Priority
TC-001	Register a new patient account	User registration	Open the registration screen, enter valid information, and click Register.	The account is created successfully, and the user is redirected to the login page.	High
TC-002	Register with missing required fields	User registration validation	Leave one or more required fields blank and click Register.	Validation messages appear, and the account is not created.	High

Test Case ID	Description	Requirements Trace	Directions	Expected Output	Priority
TC-003	Log in with valid credentials	User login	Open the login page, enter a valid email and password, and click Login.	The dashboard is displayed successfully.	High
TC-004	Log in with an incorrect password	User login validation	Enter a valid email with an incorrect password and click Login.	Login is denied, and an error message is displayed.	Medium

Test Cases (White Box)

Have a table for the test cases needed to test the User Story. A sample is below.

Test Case ID	Description	Test Steps	Expected Output
TC-101	Login	1. Go to login page 2. Enter valid username and password 3. Click "Login"	User is successfully logged in and redirected to dashboard.
TC-102	1. Add item to cart	1. Navigate to product page 2. Select a product 3. Click login	Product appears in cart with correct quantity

3.3.1 Appointment Management

Organize the test cases using the table below. Be sure to have a directory in your repository for your test suite that follows a naming scheme that matches the items in the table below.

Test Case ID	Feature Name	Test Steps	Expected Result	Priority	Notes
3.3.1	View appointment details	Log in, navigate to Appointments, and select an existing appointment.	Appointment details (provider, date, time, location) display correctly.	High	Sprint 2
3.3.2	Edit an existing appointment	Open an appointment, change the date/time, and save.	The appointment updates with the new date and time and the change is confirmed to the user.	High	Sprint 2
3.3.3	Cancel an appointment	Open an appointment, select Cancel, and confirm.	The appointment is removed from the upcoming list and a cancellation confirmation is shown.	High	Sprint 1/2
3.3.4	Reschedule an appointment	Open an appointment, select Reschedule, choose a new date/time, and confirm.	The appointment reflects the new date/time and the original time slot is released.	High	Sprint 2
3.3.5	View appointment history	Navigate to the Appointment History tab.	A list of past appointments displays with correct dates and provider details.	Medium	Sprint 2
3.3.6	View provider information from an appointment	Open an appointment and select the provider's name.	The provider's profile page opens with accurate contact and specialty information.	Medium	Sprint 2
3.3.7	Attempt to reschedule to a past date	Open an appointment, select a date earlier than today, and try to save.	A validation error displays and the appointment is not updated.	Medium	Sprint 3

Test Case ID	Feature Name	Test Steps	Expected Result	Priority	Notes
3.3.8	Navigate between appointment-related screens	From Appointments, move to History, then Provider Details, then back to Appointments.	All navigation links work correctly with no broken links or dead ends.	Medium	Sprint 3

5 Test Environment

Testing for the MEDBOOK Appointment System will be conducted using the Figma prototype. Team members will access the prototype through supported web browsers on desktop and mobile devices. Testing will focus on user registration, login, provider search, appointment scheduling, appointment management, and overall navigation. Sample patient accounts will be used during testing.

6 Milestones / Deliverables

6.1 Test Schedule

Task Name	Start	Finish	Effort	Comments
Test Planning	07/06/2026	07/13/2026	1 week	Draft and finalize this test plan document, due at the Monday 7/13 class.
Review Requirements documents	07/06/2026	07/13/2026	1 week	Cross-check SRS for updated assumptions and risks.
Create initial test estimates	07/13/2026	07/13/2026	In class	Estimate effort for Sprint 3-5 testing tasks.
Learn new test resources	07/13/2026	07/20/2026	1 week	Team reviews manual testing techniques for Figma prototypes.
First deploy to QA test environment	N/A	N/A	N/A	Not applicable at prototype stage; Figma link shared with the team instead of a QA deployment.
Functional testing - Sprint 1	06/15/2026	06/29/2026	2 weeks	Completed. Registration, login, provider search, and appointment booking verified. (Weeks 1-2, class days 6/15 & 6/22)
Iteration 2 deploy to QA test environment	N/A	N/A	N/A	Not applicable at prototype stage.
Functional testing - Sprint 2	06/29/2026	07/13/2026	2 weeks	Completed. Provider info, appointment details, edit, and history features verified. (Weeks 3-4, class days 6/29 & 7/6)
System testing	07/13/2026	07/27/2026	2 weeks	Sprint 3. Full appointment workflow tested across patient and administrator screens. (Weeks 5-6, class days 7/13 & 7/20)
Regression testing	07/27/2026	08/03/2026	1 week	Sprint 4, first half. Confirm Sprint 3 fixes did not break existing features. (Week 7, class day 7/27)
Usability Testing	08/03/2026	08/10/2026	1 week	Sprint 4, second half. Navigation flow and layout consistency review. (Week 8, class day 8/3)

Task Name	Start	Finish	Effort	Comments
Resolution of final defects and final build testing	08/10/2026	08/20/2026	1.5 weeks	Sprint 5. Address defects found during UAT ahead of the final project due Thursday 8/20. (Week 9 class day 8/10, plus buffer days before the deadline)
Deploy to Staging environment	N/A	N/A	N/A	Not applicable. MEDBOOK remains a Figma prototype for this course project.
Performance testing	N/A	N/A	N/A	Not applicable. No backend or executable code to load-test at this stage.
Release to Production	N/A	N/A	N/A	Not applicable. Final deliverable is the validated prototype, not a production release.
Final presentation / last day of semester	08/20/2026	08/20/2026	In class	Last day of the semester; final prototype presented/submitted.

6.2 Deliverables

The following deliverables will be produced as part of the MEDBOOK test effort:

- Test Plan document (this document), reviewed and updated at the start of each sprint.
- Test case documentation for each feature area, covering both black box and white box cases.
- Sprint 1 through Sprint 5 test execution results, including pass/fail status for each test case.
- A defect log tracking issues found during testing, their severity, and resolution status.
- A final prototype validation summary confirming the appointment workflow, role-based screens, and navigation are complete and error-free before submission.